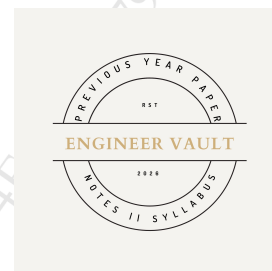


Total No. of Printed Pages: 2

SUBJECT CODE NO: - RNP-8002
FACULTY OF SCIENCE AND TECHNOLOGY
F. E. (NEP) [Group A & B]Mech, Civil, Comp. AI & ML,EEE
Examination January/February 2025
Engineering Physics

**[Time: 3:00 Hours]****[Max. Marks: 60]**

N. B

Please check whether you have got the right question paper.

- 1) Use of non-programmable calculator is allowed
- 2) Q.no.1 is compulsory
- 3) Solve any four questions from Q. no's 2, 3, 4, 5, 6 and 7

SECTION A**Q1** Solve the following questions**20**

1. The energy of ultrasonic waves depends on:
 - a) Frequency of the wave
 - b) Amplitude of the wave
 - c) Both frequency and amplitude
 - d) Only the density of the medium
2. Which of the following devices is used to generate ultrasonic waves?
 - a) Loudspeaker
 - b) piezoelectric crystal
 - c) Microphone
 - d) Magnetron
3. The phenomenon where sound waves are absorbed by the medium and do not travel further is called-
 - a) Reflection
 - b) Diffraction
 - c) Absorption
 - d) Transmission
4. What is the primary component of an X-ray tube that generates X-rays?
 - a) Reflection
 - b) Diffraction
 - c) Absorption
 - d) Transmission
5. Ultrasonic waves have frequencies:
 - a) Below 20 Hz
 - b) Between 20 Hz and 20 kHz
 - c) Above 20 kHz
 - d) Between 1 Hz and 1 kHz
6. The accelerating potential for an X-ray is 40 kV. What should be the minimum wavelength of the X-ray?
 - a) 0.024 nm
 - b) 0.031 nm
 - c) 0.097 nm
 - d) 0.054 nm
7. Which of the following is ideal for the generation of laser pulse?
 - a) Ruby laser
 - b) Carbon dioxide laser
 - c) Helium neon laser
 - d) Nd-YAG laser

- ## SECTION B

2